

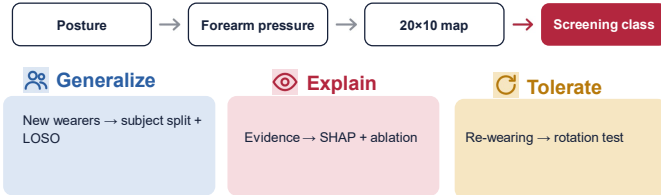
Explainable Postural Screening Using Wearable Textile Pressure Sensing

Md Sadib Rahman Ananta & Limon Bin Hossain

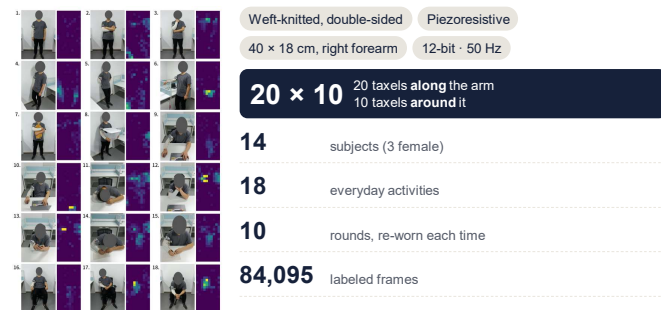
Industrial and Production Engineering · BUET, Dhaka-1000, Bangladesh



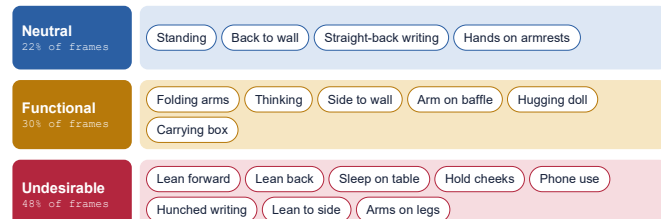
Why a pressure sleeve?



The Smart-Sleeve



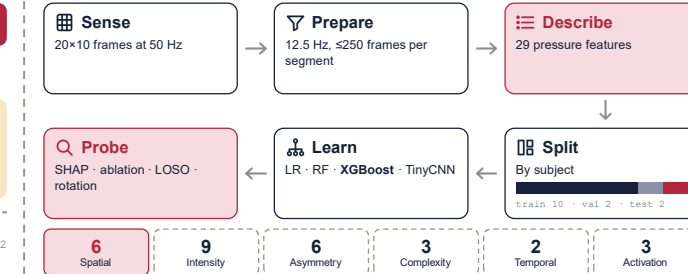
18 activities → 3 classes



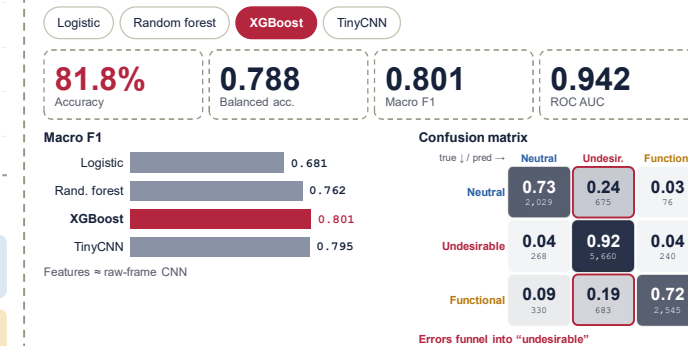
Other mappings, same result



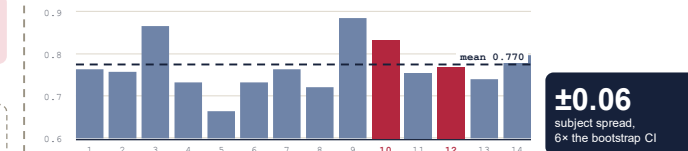
Pipeline



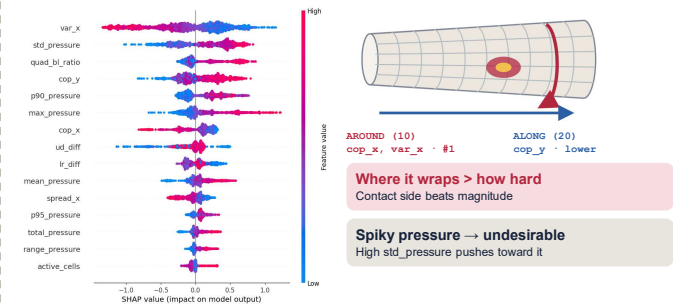
Results on unseen wearers



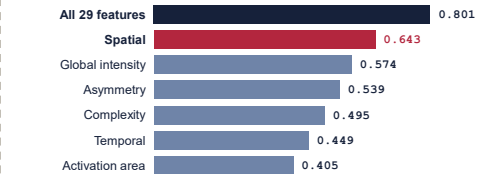
Leave-one-subject-out



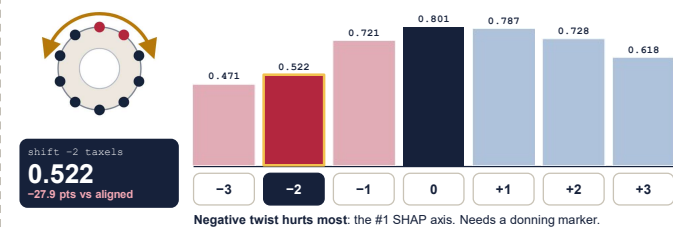
What the sleeve "feels"



One feature family at a time · macro F1



Re-wearing stress test



Take-home **0.80** macro F1 on unseen wearers

Around-the-arm location beats pressure size

8 undesirable activities, recall 0.74–1.00

REFERENCE

G. Xu, W. Deng, J. Cheng et al., Smart-Sleeve dataset, *Sensors* 22(5), 1702, 2022.

github.com/xghgithub/Smart-Sleeve-Dataset